

DATASET PROFILING & SUMMARY STAT

1. Dataset Overview

The dataset used in this study is derived from the **Adult Psychiatric Morbidity Survey (APMS 2014)** and contains statistical information related to mental health conditions across different demographic groups. It primarily represents the distribution of mental health indicators across various age categories and gender groups.

The dataset consists of a total of **24 observations and 10 variables**. The variables include one categorical column representing the **type of mental health category**, another categorical column indicating the **population group (Men, Women, and All Adults)**, and multiple numerical columns representing **percentage values across different age groups**.

The numerical variables are stored in floating-point format and reflect the proportion or percentage of individuals affected within each demographic segment. The categorical variables are stored as object data types and are used to classify and group the data for analysis.

During the initial data profiling, it was observed that the dataset contains a **moderate proportion of missing values (approximately 20.83%)** in most of the numerical columns. However, no duplicate records were identified, ensuring the uniqueness of each observation.

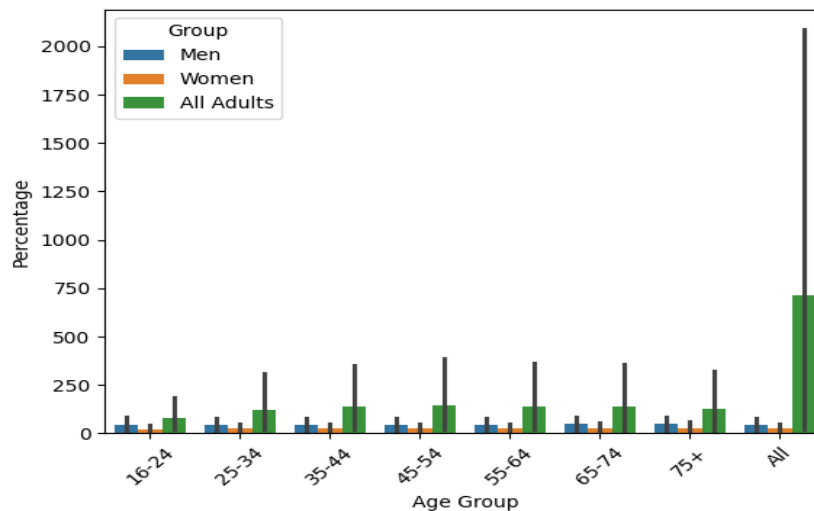
2. Data Types

The dataset comprises both **categorical** and **numerical** variables, each serving a distinct role in the analysis.

The **categorical variables** include *Category* and *Group*, which are stored as object data types. The *Category* column represents different mental health conditions or classifications, while the *Group* column categorizes the population into segments such as Men, Women, and All Adults. These variables are essential for grouping and comparative analysis.

The **numerical variables** consist of age-based columns, including 16–24, 25–34, 35–44, 45–54, 55–64, 65–74, 75+, and All. These columns are stored as floating-point data types and represent percentage values corresponding to each demographic segment. The use of numeric data types allows for statistical computations such as mean, standard deviation, and other descriptive measures.

- **Category, Group → Object (Categorical)**
- **All age columns → Numeric (float/int)**



3. Missing Values Analysis

The dataset was examined to identify the presence and distribution of missing values across all variables. Missing data is an important aspect of data quality, as it can influence the reliability and accuracy of analytical results.

The analysis reveals that the **categorical variables**, namely Category and Group, do not contain any missing values, indicating complete information for classification and grouping purposes.

However, the **numerical variables** corresponding to age groups (16–24, 25–34, 35–44, 45–54, 55–64, 65–74, 75+, and All) exhibit a consistent pattern of missing data. Each of these columns contains **5 missing values out of 24 total observations**, resulting in approximately **20.83% missing data per column**.

This level of missingness can be considered **moderate** and may impact statistical analysis if not handled appropriately. The uniform distribution of missing values across numerical columns suggests that the missingness may not be random and could be associated with specific categories or groups within the dataset.

To ensure data integrity, appropriate handling techniques such as **data imputation (mean/median substitution)** or **removal of incomplete records** may be applied depending on the analysis requirements. Careful treatment of missing values is essential to maintain the validity and robustness of the results.

Check Missing Value

```
missing_percent = (df.isnull().sum() / len(df)) * 100
print("\nMissing percentage:")
print(missing_percent)
```

```
Missing percentage:
Category      0.000000
16-24         20.833333
25-34         20.833333
35-44         20.833333
45-54         20.833333
55-64         20.833333
65-74         20.833333
75+           20.833333
All           20.833333
Group         0.000000
dtype: float64
```

4. Duplicate Records Analysis

The dataset was analyzed to identify the presence of duplicate records, which can negatively impact data quality by introducing redundancy and bias in the analysis.

A duplicate check was performed using standard data profiling techniques, where each row was compared across all columns to detect exact repetitions. The results indicate that **no duplicate records were found in the dataset**.

The absence of duplicate entries confirms that each observation in the dataset is **unique and independently recorded**, ensuring the integrity and reliability of the data. This eliminates the need for any duplicate removal or further preprocessing in this regard.

Duplicate Check

```
duplicates = df.duplicated().sum()
print("\nNumber of duplicate rows:", duplicates)
```

```
Number of duplicate rows: 0
```

5. Descriptive Statistics

Descriptive statistical analysis was performed on the numerical variables of the dataset to summarize the central tendency, dispersion, and overall distribution of the data. This analysis provides a comprehensive understanding of the patterns and variability present across different age groups.

The key statistical measures considered include **count, mean, standard deviation, minimum, maximum, and quartiles (25%, 50%, and 75%)** for each numerical column. The results indicate that the **mean values vary**

significantly across age groups, reflecting differences in the distribution of mental health indicators among different demographic segments.

The **standard deviation values are relatively high** for several variables, suggesting a considerable spread in the data. This indicates that the observations are not closely clustered around the mean and that there is substantial variability within the dataset.

Furthermore, the presence of **large maximum values in comparison to the mean and median** highlights the possibility of extreme values or outliers in the dataset. Such values can influence the overall statistical measures and may require further investigation.

The quartile analysis shows the distribution of data across different ranges, with the median (50th percentile) providing a measure of central tendency that is less affected by extreme values. The gap between the lower (25%) and upper (75%) quartiles indicates the interquartile range, which reflects the spread of the middle portion of the data.

Descriptive Statistics

```
print(df.describe(include='all'))
```

	Category	16-24	25-34	35-44	45-54	\
count	24	19.000000	19.000000	19.000000	19.000000	
unique	12	NaN	NaN	NaN	NaN	
top	0-5	NaN	NaN	NaN	NaN	
freq	3	NaN	NaN	NaN	NaN	
mean	NaN	61.047368	86.052632	93.684211	99.684211	
std	NaN	124.572908	231.728573	264.717827	290.701575	
min	NaN	4.200000	7.400000	6.800000	6.200000	
25%	NaN	10.200000	9.750000	10.500000	11.350000	
50%	NaN	16.700000	17.200000	16.200000	18.000000	
75%	NaN	70.550000	69.300000	68.050000	68.700000	
max	NaN	560.000000	1035.000000	1180.000000	1294.000000	

	55-64	65-74	75+	All	Group
count	19.000000	19.000000	19.000000	19.000000	24
unique	NaN	NaN	NaN	NaN	3
top	NaN	NaN	NaN	NaN	All Adults
freq	NaN	NaN	NaN	NaN	16
mean	96.105263	94.163158	87.468421	428.742105	NaN
std	275.240681	267.521750	238.885084	1723.803891	NaN
min	5.800000	3.600000	1.100000	5.800000	NaN
25%	9.550000	6.600000	5.000000	9.550000	NaN
50%	17.100000	12.900000	12.800000	15.700000	NaN
75%	68.100000	78.950000	79.750000	71.300000	NaN
max	1226.000000	1189.000000	1062.000000	7546.000000	NaN

6. Data Transformation

Data transformation was performed to convert the dataset into a suitable format for analysis and visualization. This step is essential for improving data usability, consistency, and interpretability.

Initially, the dataset was structured in a **wide format**, where multiple age group columns (such as 16–24, 25–34, etc.) were represented as separate variables. While this format is useful for storage, it is not ideal for analytical operations and visualization.

To address this, the dataset was transformed into a **long format** using a reshaping technique. Specifically, the *melt* function was applied to convert multiple age group columns into two columns: one representing the **Age Group** and the other representing the corresponding **Percentage values**. The Category and Group columns were retained as identifier variables during this transformation.

This transformation resulted in a more flexible and analysis-friendly structure, where each row represents a single observation for a specific category, group, and age segment. The long format is particularly advantageous for data visualization, as it allows for easier comparison across different groups and categories using plots such as bar charts and box plots.

7. Outlier Detection

Outlier detection was conducted to identify extreme values in the dataset that deviate significantly from the general pattern of the data. Detecting outliers is essential, as they can influence statistical measures and potentially distort analytical results.

In this study, outliers were identified using **box plot visualization** and statistical interpretation of descriptive measures such as minimum, maximum, and quartile values. Box plots provide a clear representation of the data distribution and help in detecting observations that fall outside the typical range.

The analysis reveals the presence of **extremely high maximum values** in several numerical columns when compared to their respective mean and median values. These unusually large values indicate potential outliers or inconsistencies in the dataset. For instance, certain age group columns exhibit values that are disproportionately higher than the majority of observations.

Such outliers may arise due to various reasons, including **data entry errors, differences in scaling, or genuine extreme observations** within the population.

The presence of these values contributes to the high standard deviation observed in the descriptive statistics, further confirming data variability.

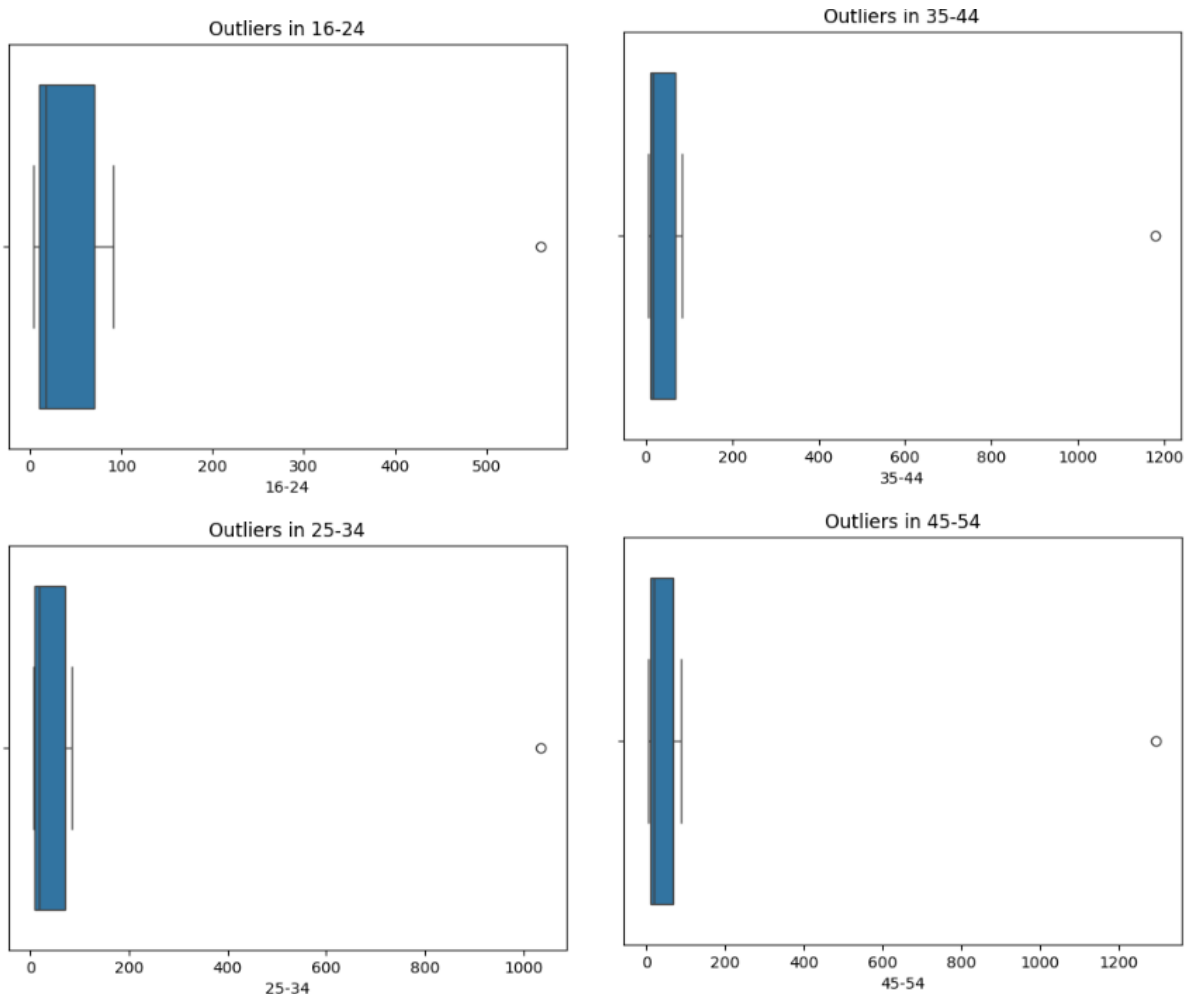
It is important to handle outliers carefully to maintain the integrity of the analysis. Depending on the context, appropriate techniques such as **outlier removal**, **transformation**, or **capping** may be applied. However, in cases where outliers represent real-world phenomena, they should be retained and interpreted with caution.

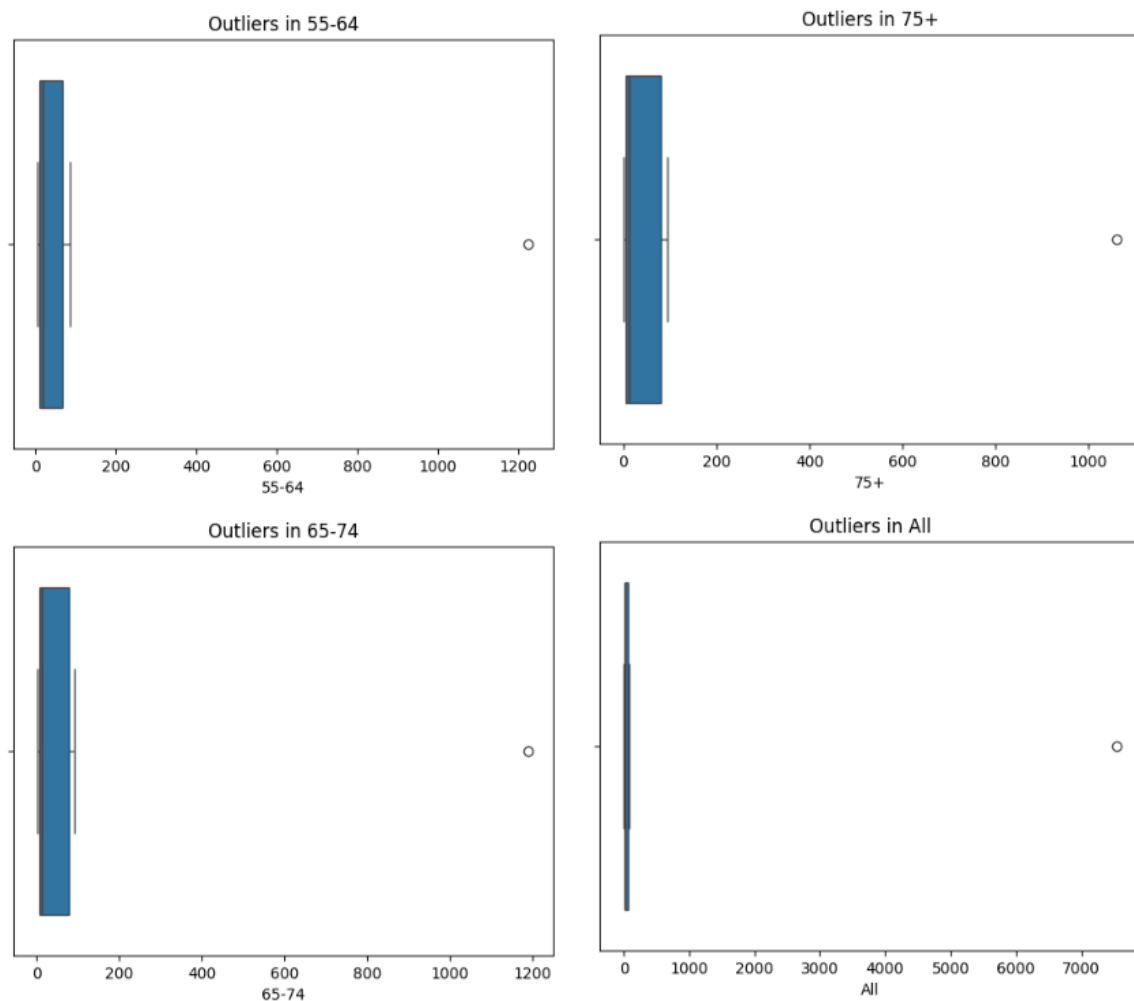
Outlier Detection

```
import seaborn as sns
import matplotlib.pyplot as plt

num_cols = ["16-24", "25-34", "35-44", "45-54", "55-64", "65-74", "75+", "All"]

for col in num_cols:
    plt.figure()
    sns.boxplot(x=df[col])
    plt.title(f"Outliers in {col}")
    plt.show()
```





8. Profiling Summary

A comprehensive data profiling process was conducted to evaluate the structure, quality, and characteristics of the dataset. The profiling results provide insights into data types, missing values, and the uniqueness of each variable.

The dataset consists of both **categorical and numerical variables**. The Category and Group columns are categorical in nature and do not contain any missing values, indicating complete and reliable classification information. The Category column includes **12 unique values**, while the Group column consists of **3 distinct categories**, representing different population segments.

The remaining columns correspond to **age-based numerical variables**, all stored as floating-point data types. Each of these columns contains **19 unique values**, suggesting a high level of variability and minimal repetition in the data. However, these numerical columns also exhibit **5 missing values each**, accounting for approximately **20.83% missing data per column**.

The presence of consistent missing values across all numerical columns indicates a uniform pattern of data incompleteness. While this level of missingness is moderate, it requires careful handling to ensure accurate analysis.

```
summary = pd.DataFrame({
    "Column": df.columns,
    "Data Type": df.dtypes,
    "Missing Values": df.isnull().sum(),
    "Missing %": (df.isnull().sum() / len(df)) * 100,
    "Unique Values": df.nunique()
})

print(summary)
```

	Column	Data Type	Missing Values	Missing %	Unique Values
Category	Category	object	0	0.000000	12
16-24	16-24	float64	5	20.833333	19
25-34	25-34	float64	5	20.833333	19
35-44	35-44	float64	5	20.833333	19
45-54	45-54	float64	5	20.833333	19
55-64	55-64	float64	5	20.833333	19
65-74	65-74	float64	5	20.833333	19
75+	75+	float64	5	20.833333	19
All	All	float64	5	20.833333	18
Group	Group	object	0	0.000000	3